

35.4 Goodness of fit

In the normal linear model the residual sum of squares both estimates σ^2 and measures fit, and its known distribution makes a fit too poor to be chance announce itself (Chapter 14). Binary data have no free dispersion parameter, so the deviance has only the second role, and often cannot fill it either.

35.4.1 Two statistics for grouped data

Definition 35.4.1 — Pearson and deviance statistics

For grouped binomial data $y_i \sim \text{Binomial}(m_i, \pi_i)$, $i = 1, \dots, N$, with fitted probabilities $\hat{\pi}_i$ from a model with p parameters, the **Pearson statistic** and the **deviance** are

$$\begin{aligned} X^2 &= \sum_{i=1}^N \frac{(y_i - m_i \hat{\pi}_i)^2}{m_i \hat{\pi}_i (1 - \hat{\pi}_i)}, \\ D &= 2 \sum_{i=1}^N \left\{ y_i \log \frac{y_i}{m_i \hat{\pi}_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i - m_i \hat{\pi}_i} \right\}, \end{aligned} \quad (35.4.1)$$

with the convention $0 \log 0 = 0$. Both compare the fit with the **saturated** model that gives each group its own probability.

X^2 is the sum over all $2N$ cells of successes and failures of $(\text{observed} - \text{fitted})^2 / \text{fitted}$, and D is twice the sum of $\text{observed} \times \log(\text{observed}/\text{fitted})$: the classical categorical-data statistics, specialized to the binomial.

Proposition 35.4.1 — What the two statistics measure

- (a) (*Grouped data.*) Fix N and the model, let $m_i \rightarrow \infty$ for every i with π_i bounded away from 0 and 1, and suppose the model holds. Then X^2 and D both converge in distribution to $\chi^2(N - p)$, and $X^2 - D \rightarrow 0$ in probability.
- (b) (*Ungrouped data, canonical link.*) If $m_i = 1$ for every i and $\hat{\beta}$ solves the logistic likelihood equations Eq. 35.1.2, then

$$D = -2 \sum_{i=1}^n \{ \hat{\pi}_i \log \hat{\pi}_i + (1 - \hat{\pi}_i) \log(1 - \hat{\pi}_i) \}. \quad (35.4.2)$$

The deviance is therefore a function of the fitted probabilities alone: it does not depend on which observations were successes, it is bounded above by $2n \log 2$, and it has no $\chi^2(n - p)$ limit.

- (c) (*Differences.*) For nested models $\omega \subset \Omega$ with q parameters between them, $D_\omega - D_\Omega$ does not depend on the grouping and converges to $\chi^2(q)$ under the smaller model, grouped or not.

Proof.

- (a) The statement about D is Theorem 34.5.5 specialized to the binomial family: with N fixed and every $m_i \rightarrow \infty$, the model is a smooth submodel of an N -parameter family whose maximum likelihood estimates are asymptotically normal, so the likelihood ratio statistic against the saturated model has $N - p$ degrees of freedom. For X^2 , put $b = m_i \hat{\pi}_i$ and $r_i = (y_i - b) / \{m_i \hat{\pi}_i (1 - \hat{\pi}_i)\}^{1/2}$, so that $r_i = O_p(1)$ and $y_i - b = O_p(m_i^{1/2})$. The i th deviance summand is $g(y_i)$ for $g(a) = 2\{a \log(a/b) + (m_i - a) \log((m_i - a)/(m_i - b))\}$, and $g(b) = g'(b) = 0$ and $g''(b) = 2/\{m_i \hat{\pi}_i (1 - \hat{\pi}_i)\}$, while the third derivative is $O(m_i^{-2})$ near that point. Taylor's theorem therefore gives $g(y_i) = r_i^2 + O_p(m_i^{-1/2})$, using $\hat{\pi}_i$ bounded away from 0 and 1. Summing over the N fixed groups, $D - X^2 \rightarrow 0$ in probability, so X^2 has the same limit as D .
- (b) Write $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}$, so that $\hat{\eta}_i = \log \hat{\pi}_i - \log(1 - \hat{\pi}_i)$. For ungrouped data the saturated model sets $\hat{\pi}_i = y_i$ and has log-likelihood zero, so

$$\begin{aligned} D &= -2 \sum_i \{y_i \log \hat{\pi}_i + (1 - y_i) \log(1 - \hat{\pi}_i)\} \\ &= -2 \sum_i \{y_i \hat{\eta}_i + \log(1 - \hat{\pi}_i)\} \\ &= -2 \left\{ \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} + \sum_i \log(1 - \hat{\pi}_i) \right\}. \end{aligned}$$

The likelihood equations say $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \hat{\boldsymbol{\pi}}$, so $\mathbf{X}^\top \mathbf{y}$ may be replaced by $\mathbf{X}^\top \hat{\boldsymbol{\pi}}$, giving $D = -2 \sum_i \{\hat{\pi}_i \hat{\eta}_i + \log(1 - \hat{\pi}_i)\}$, which is Eq. 35.4.2 after substituting for $\hat{\eta}_i$. Each summand of Eq. 35.4.2 is at most $2 \log 2$, attained at $\hat{\pi}_i = 1/2$, which gives the bound. For the last claim write $H(\pi) = -\{\pi \log \pi + (1 - \pi) \log(1 - \pi)\}$ and $f_n(\boldsymbol{\beta}) = 2 \sum_i H(\pi_i)$ with $\pi_i = \pi_i(\boldsymbol{\beta})$, so that $D = f_n(\hat{\boldsymbol{\beta}})$ identically: D is a smooth deterministic function of $\hat{\boldsymbol{\beta}}$, with no other dependence on the data. Now $dH/d\eta = -\eta \pi(1 - \pi)$, and $\pi(1 - \pi) \leq e^{-|\eta|}$, so $|dH/d\eta| \leq |\eta| e^{-|\eta|} \leq e^{-1}$ and $\|\nabla f_n\| \leq 2e^{-1} \sum_i \|\mathbf{x}_{(i)}\| = O(n)$ for bounded covariates. With $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = O_p(n^{-1/2})$ from Theorem 34.5.1, the mean value theorem gives $D - f_n(\boldsymbol{\beta}) = O_p(n^{1/2})$, so

$$n^{-1}D - 2n^{-1} \sum_{i=1}^n H(\pi_i) \rightarrow 0 \text{ in probability,}$$

whereas $n^{-1}\chi^2(n - p) \rightarrow 1$. The two agree only by accident: the simulation of Example (Three deviances that mean nothing) has $n^{-1}D$ averaging $206.8/200 = 1.03$ under one correct model and $263.6/200 = 1.32$ under another. So D has no $\chi^2(n - p)$ limit. The same bound leaves D fluctuating about $f_n(\boldsymbol{\beta})$ by $O_p(n^{1/2})$, which is the order of the standard deviations in Example (Three deviances that mean nothing); what fails is not the spread of D but its location.

- (c) The saturated log-likelihood is common to both deviances and cancels, so the difference is twice a log-likelihood ratio between two nested smooth models with q free parameters between them, and Theorem 34.5.5 applies with q fixed.

■

Example. (Is the party scale linear on the logit scale?)

Collapse the survey of Example (Voting intention in the 1996 American election) onto its seven party groups, of between 37 and 200 respondents each, and fit $\text{logit } \pi = \beta_0 + \beta_1 \text{ party}$. The deviance is 12.765 on 5 degrees of freedom ($p = 0.0257$) and the Pearson statistic is 11.899 ($p = 0.0362$). The Pearson residuals locate the misfit: -2.067 at “independent-Democrat”, where the model predicts 0.132 against an observed 0.065, and 2.122 at “independent-Republican”, where it predicts 0.640 against 0.745. The transition across the middle of the scale is sharper than a straight line in the log odds allows, and a quadratic term does not help: the deviance falls only to 12.762, so the misfit is a step, not a curve.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from scipy import stats

anes = sm.datasets.anes96.load_pandas().data
cells = anes.groupby("PID")["vote"].agg(["sum", "count"]).reset_index()
counts = np.column_stack([cells["sum"], cells["count"] - cells["sum"]])
Xg = sm.add_constant(cells[["PID"]])

g = sm.GLM(counts, Xg, family=sm.families.Binomial()).fit()
print(cells.to_string(index=False))
print(f"deviance {g.deviance:.3f} on {g.df_resid} df, p = {stats.chi2.sf(g.deviance,
    g.df_resid):.4f}")
print(f"Pearson {g.pearson_chi2:.3f} on {g.df_resid} df, "
    f"p = {stats.chi2.sf(g.pearson_chi2, g.df_resid):.4f}")
```

[Run this code online](#)

The conditions in part (a) are not decoration. If N grows with n instead, as it does whenever a continuous covariate is in the model, the approximation fails, worst of all in the extreme case $m_i \equiv 1$ of part (b).

35.4.2 The ungrouped deviance measures nothing

Eq. 35.4.2 deserves to be read twice. The left-hand side looks like a discrepancy between data and fit; the right-hand side contains no data at all. It is twice the total Bernoulli entropy of the fitted probabilities, a measure of how *undecided* the model is: probabilities near $1/2$ give a deviance near $2n \log 2$ and extreme ones give a small deviance, whether or not the model is correct.

Example. (Three deviances that mean nothing)

Simulate $n = 200$ observations with one covariate and an intercept in three ways: a correct model with a strong slope, a correct model with a weak slope, and a model that leaves a genuine quadratic effect out of the fitted predictor. All three fit two parameters, so $n - p = 198$ and a chi-squared reference would have mean 198 and standard deviation 19.9. Over 2000 replications:

data-generating mechanism	mean deviance	standard deviation
correct, strong slope	206.8	14.3

correct, weak slope	263.6	7.0
quadratic omitted	252.2	8.5

Two *correct* models with the same n and p differ by nearly sixty in mean deviance, and the badly misspecified model has a *smaller* deviance than one of them: no reference distribution can serve all three (Figure 35.4.1).

The survey model of Example (Voting intention in the 1996 American election) has ungrouped deviance 528.29 on 939 degrees of freedom; dropping every covariate but education gives an obviously wrong model with deviance 1275.16 on 942. Neither number can be referred to a table.

```
# the ungrouped deviance is a function of the fitted probabilities alone
X = pd.DataFrame({"const": 1.0, "PID": anes["PID"], "age": anes["age"] / 10,
                  "educ": anes["educ"], "income": anes["income"]})
y = anes["vote"].to_numpy()
fit = sm.GLM(y, X, family=sm.families.Binomial()).fit()
p = fit.fittedvalues.to_numpy()

entropy = -2 * np.sum(p * np.log(p) + (1 - p) * np.log(1 - p))
print("deviance reported ", round(fit.deviance, 6))
print("entropy of the fit ", round(entropy, 6))
```

[Run this code online](#)

```
def deviance_null(n, B, seed=0):
    """Ungrouped deviances under three data-generating mechanisms: a correct model
    with a
    strong slope, a correct model with a weak slope, and a model that omits a real
    quadratic
    effect. All three fit an intercept and one slope to n observations."""
    rng = np.random.default_rng(seed)
    out = np.empty((B, 3))
    for b in range(B):
        x = rng.normal(size=n)
        Z = np.column_stack([np.ones(n), x])
        for j, eta in enumerate([0.4 + 1.5 * x, 0.4 + 0.3 * x, 0.4 + 1.5 * x - 1.8 *
                                x ** 2]):
            yb = (rng.random(n) < 1 / (1 + np.exp(-eta))).astype(float)
            out[b, j] = sm.GLM(yb, Z, family=sm.families.Binomial()).fit().deviance
    return out

demo = deviance_null(200, 100, seed=1)
print("n = 200 and p = 2, so a chi-squared reference would have mean 198 and "
      f"standard deviation {np.sqrt(2 * 198):.1f}")
for j, label in enumerate(["correct, strong slope", "correct, weak slope ",
                           "quadratic omitted "]):
    print(f"{label}: mean {demo[:, j].mean():6.1f}, "
          f"standard deviation {demo[:, j].std():4.1f}")
```

[Run this code online](#)

The Pearson statistic escapes Eq. 35.4.2, since $X^2 = \sum_i \{y_i / \hat{\pi}_i + (1 - y_i) / (1 - \hat{\pi}_i)\} - n$ does depend on y , but gains nothing by it (Exercise (B1)). What survives is part (c): for ungrouped

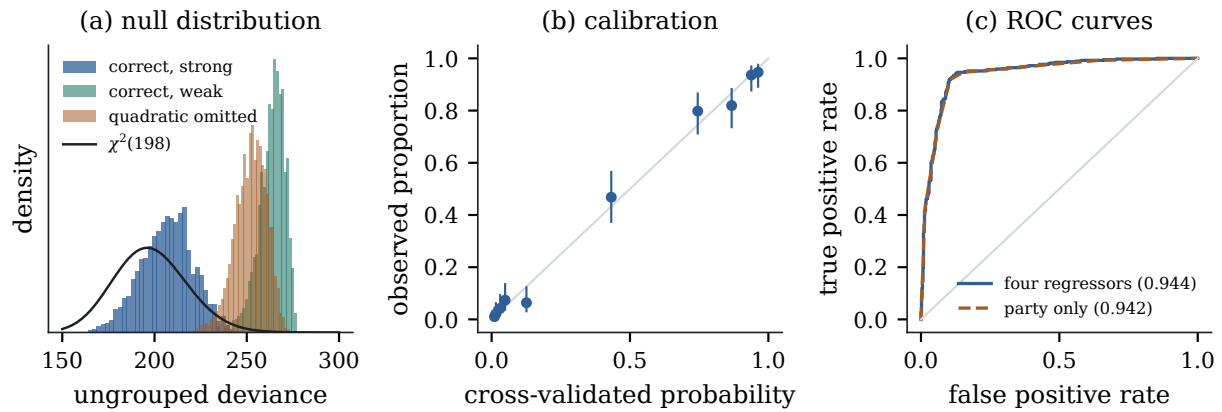


Figure 35.4.1: (a) Simulated null distributions of the ungrouped deviance with $n = 200$ and two fitted parameters, under the three mechanisms of [Example \(Three deviances that mean nothing\)](#), against the $\chi^2(198)$ density. (b) Cross-validated calibration of the survey model in ten bins, with pointwise intervals. (c) ROC curves for the four-regressor model and for party identification alone.

data, test the model against enlargements of it rather than against the saturated alternative, and use AIC or cross-validation ([Chapter 29](#)) when the comparisons are not nested.

35.4.3 Grouping by fitted values

Hosmer and Lemeshow (1980) proposed making grouped data out of ungrouped: sort by $\hat{\pi}_i$, cut into G groups of roughly equal size, and compare

$$\hat{C} = \sum_{k=1}^G \frac{(O_k - E_k)^2}{E_k(1 - E_k/n_k)}, \quad O_k = \sum_{i \in k} y_i, \quad E_k = \sum_{i \in k} \hat{\pi}_i,$$

referred to $\chi^2(G - 2)$. The reference distribution has no proof behind it: it was established by simulation, and the two lost degrees of freedom are an empirical finding. The statistic is a useful descriptive check and an unstable test.

Example. (The same model, three verdicts)

For the four-regressor model of [Example \(Voting intention in the 1996 American election\)](#):

groups G	$\hat{C}$	p
8	8.670	0.193
10	12.232	0.141
12	16.119	0.096

The p -value halves as G goes from 8 to 12, and different software makes different choices about G , about how to break ties in $\hat{\pi}_i$, and about whether to use equal-sized groups or equally spaced cut-points. None of these choices is more principled than any other.

```
def hosmer_lemeshow(y, p, groups):
```

```

"""The Hosmer-Lemeshow statistic and its nominal p-value with the given number
of groups."""
order = np.argsort(p)
chunks = np.array_split(order, groups)
stat = 0.0
for c in chunks:
    e = p[c].sum()
    stat += (y[c].sum() - e) ** 2 / (e * (1 - e / len(c)))
return stat, stats.chi2.sf(stat, groups - 2)

for gr in (8, 10, 12):
    stat, pv = hosmer_lemeshow(y, p, gr)
    print(f"{gr:2d} groups: statistic {stat:6.3f}, p = {pv:.3f}")

```

[Run this code online](#)

35.4.4 Calibration and discrimination

Two questions hide inside “does the model fit”. **Calibration** asks whether the fitted probabilities are right: among observations with $\hat{\pi}_i \approx 0.3$, do about thirty per cent have $y_i = 1$? **Discrimination** asks whether they order the observations correctly, and a model can have either without the other. Classification tables of sensitivity against specificity answer a third question, and depend on a threshold and on a loss they rarely state; pseudo- R^2 measures such as $1 - D/D_{\text{null}}$ have no interpretation of their own, and by Proposition 35.4.1(b) the ungrouped version of that ratio is a ratio of entropies.

Proposition 35.4.2 — In-sample calibration is automatic

Let $\hat{\beta}$ solve Eq. 35.1.2 with $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, let $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\beta}$, and suppose $\hat{\eta}$ is not constant. Fit the two-parameter logistic model $\text{logit } \pi_i = a + b\hat{\eta}_i$ to the same responses. Then $\hat{a} = 0$ and $\hat{b} = 1$ exactly.

Proof. The likelihood equations of the two-parameter model at $(a, b) = (0, 1)$ are $\sum_i (y_i - \hat{\pi}_i) = 0$ and $\sum_i \hat{\eta}_i (y_i - \hat{\pi}_i) = 0$. The first holds because $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ and $\mathbf{X}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = \mathbf{0}$; the second because $\hat{\boldsymbol{\eta}} = \mathbf{X}\hat{\boldsymbol{\beta}} \in \mathcal{C}(\mathbf{X})$, so $\hat{\boldsymbol{\eta}}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = 0$. So $(0, 1)$ solves the two-parameter likelihood equations; it remains to see that a maximizer exists. The design $[\mathbf{1}, \hat{\boldsymbol{\eta}}]$ has rank two, since $\hat{\eta}$ is not constant. Suppose some $(a, b) \neq (0, 1)$ had $s_i(a + b\hat{\eta}_i) \geq 0$ for every i . As $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ there is an $\mathbf{e}$ with $\mathbf{X}\mathbf{e} = \mathbf{1}$, and $\mathbf{b}^* = a\mathbf{e} + b\hat{\boldsymbol{\beta}}$ satisfies $\mathbf{X}\mathbf{b}^* = a\mathbf{1} + b\hat{\boldsymbol{\eta}} \neq \mathbf{0}$, so $\mathbf{b}^* \neq \mathbf{0}$ separates $\mathbf{X}$ — contradicting the overlap that the finiteness of $\hat{\boldsymbol{\beta}}$ guarantees (Theorem 35.3.1). The two-parameter data are therefore in overlap, and Theorem 35.3.1(c) gives a unique maximizer, which must be the stationary point $(0, 1)$. ■

The **calibration slope** $\hat{b}$ is therefore worthless on the data that produced $\hat{\boldsymbol{\beta}}$. It becomes informative when $\hat{\eta}$ comes from a held-out fold or a new sample, where $\hat{b} < 1$ is the signature of overfitting. On the survey model of Example (Voting intention in the 1996 American election), ten-fold cross-validated predictions give a slope of 0.983 and an intercept of 0.005: four regressors

and 944 observations is not a regime in which overfitting bites.

Proposition 35.4.3 — What the area under the ROC curve is, and is not

Let $\hat{s}_i$ be any scores, let I be a uniformly chosen index with $y_I = 1$ and J an independent uniformly chosen index with $y_J = 0$. The area under the curve traced by (false positive rate, true positive rate) as a threshold moves through the scores equals

$$\text{AUC} = \Pr(\hat{s}_I > \hat{s}_J) + \frac{1}{2} \Pr(\hat{s}_I = \hat{s}_J),$$

the normalized Mann–Whitney statistic. Consequently the AUC depends on the scores only through their ranks, so it is unchanged by every strictly increasing transformation of $\hat{s}$, and in particular it says nothing whatever about calibration.

Proof. Order the distinct score values downwards. Moving the threshold past one of them advances the curve by the fraction ΔFPR of controls and the fraction ΔTPR of cases at that value, adding the trapezoid $\Delta\text{FPR}\{\text{TPR} + \frac{1}{2}\Delta\text{TPR}\}$. Summing, the first part counts pairs with $\hat{s}_I > \hat{s}_J$ and the second counts ties with weight $\frac{1}{2}$, each divided by $n_1 n_0$. Invariance is immediate, since an increasing transformation preserves every comparison $\hat{s}_I > \hat{s}_J$. ■

So a model whose probabilities are all wrong by a factor of ten has the same AUC as the correctly calibrated model with the same ordering. The AUC is insensitive in the other direction too: for Example (Voting intention in the 1996 American election) party identification alone gives $\text{AUC} = 0.942$, and adding age, education and income raises it to 0.944. An increment of 0.002 in a rank-based summary is the wrong scale on which to judge the three covariates.

35.4.5 Residuals and influence

The diagnostics of [Chapter 20](#) carry over, with the working weights as the metric.

Definition 35.4.2 — Residuals for binomial data

With $\hat{w}_i = m_i \hat{\pi}_i (1 - \hat{\pi}_i)$, $\hat{\mathbf{W}} = \text{diag}(\hat{w}_i)$ and $\hat{h}_i$ the i th diagonal entry of $\hat{\mathbf{W}}^{1/2} \mathbf{X} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$, the **Pearson** and **deviance** residuals are

$$r_i = \frac{y_i - m_i \hat{\pi}_i}{\sqrt{\hat{w}_i}}, \quad d_i = \text{sign}(y_i - m_i \hat{\pi}_i) \sqrt{D_i},$$

where D_i is the i th summand of the deviance in [Eq. 35.4.1](#). Their **standardized** forms divide by $(1 - \hat{h}_i)^{1/2}$.

By construction $\sum_i r_i^2 = X^2$ and $\sum_i d_i^2 = D$. The matrix in the definition is the hat matrix of the last weighted least squares step of Proposition 35.1.3, so it is symmetric and idempotent with trace $\text{rank}(\mathbf{X})$, and [Section 6.8](#) applies to it unchanged. The analogue of Cook's distance (Definition 20.4.1) is $r_i^2 \hat{h}_i / \{p(1 - \hat{h}_i)^2\}$.

Two things change. For ungrouped data r_i takes only two values once $\hat{\pi}_i$ is given, so plots against a covariate show two curves rather than a cloud and must be smoothed or binned. And $\hat{h}_i$ carries the factor $\hat{w}_i$, so high leverage now means a covariate vector that is extreme *and* a fitted probability near $1/2$; a point whose outcome the model regards as certain has low leverage wherever its covariates lie, the opposite of the linear-model intuition.

35.4.6 Exercises

35.4.6.1 A. Check your understanding

Exercise A1.

Use Eq. 35.4.2 to find the largest and smallest possible values of the ungrouped deviance of a logistic fit, and describe the fits that attain them.

Solution. Each summand $-2\{\hat{\pi} \log \hat{\pi} + (1 - \hat{\pi}) \log(1 - \hat{\pi})\}$ lies in $[0, 2 \log 2]$, so $0 \leq D \leq 2n \log 2$. The upper bound is attained when every $\hat{\pi}_i = 1/2$, as for the intercept-only model fitted to equal numbers of successes and failures. The lower bound is approached only as the fitted probabilities go to 0 and 1, which by Theorem 35.3.1 needs complete separation, so 0 is an infimum and not a minimum.

35.4.6.2 B. Practice

Exercise B1.

Show that for ungrouped data $X^2 = \sum_i \{y_i/\hat{\pi}_i + (1 - y_i)/(1 - \hat{\pi}_i)\} - n$, and deduce that for the intercept-only model $X^2 = n$ exactly, whatever the data. What does that say about X^2 as a test of fit?

Solution. With $m_i = 1$, $(y_i - \hat{\pi}_i)^2 / \{\hat{\pi}_i(1 - \hat{\pi}_i)\}$ equals $(1 - \hat{\pi}_i)/\hat{\pi}_i$ when $y_i = 1$ and $\hat{\pi}_i/(1 - \hat{\pi}_i)$ when $y_i = 0$, and both equal $y_i/\hat{\pi}_i + (1 - y_i)/(1 - \hat{\pi}_i) - 1$; summing gives the identity. For the intercept-only model $\hat{\pi}_i = \bar{y}$ for every i , so $X^2 = n\bar{y}/\bar{y} + n(1 - \bar{y})/(1 - \bar{y}) - n = n$. A statistic that equals the sample size for every data set cannot test anything.

Exercise B2.

Let $\hat{\eta}$ be fitted on one sample and evaluated on an independent sample from the same population. Show that the calibration slope b has probability limit 1 if the model is correct and $\hat{\beta}$ is consistent, and explain why a value below one indicates overfitting.

Exercise B3.

Suppose the covariate is $N(\mu_1, 1)$ among cases and $N(\mu_0, 1)$ among controls. Compute the AUC of the correctly specified logistic model in terms of $\Delta = \mu_1 - \mu_0$, and find the Δ needed for an AUC of 0.9.

Solution. The fitted score is monotone in the covariate, so the AUC is $\Pr(X_1 > X_0) = \Phi(\Delta/\sqrt{2})$, since $X_1 - X_0 \sim N(\Delta, 2)$. Setting this to 0.9 gives $\Delta = \sqrt{2} \Phi^{-1}(0.9) = 1.812$.

An AUC of 0.65 needs only $\Delta \approx 0.55$, a difference highly significant in a large sample yet nearly useless for classifying an individual.